

Curriculum for Certified Professional for Software
Architecture (CPSA)[®] Advanced Level

SWARC4AI – *Software Architecture for AI Systems*

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Learning Goals Overview

- LG 1-1: Classify artificial intelligence and machine learning, data science, deep learning, and generative AI
- LG 1-2: Identify, classify, prioritize, and evaluate AI use cases from a business perspective
- LG 1-3: Classify the strengths, limitations, and typical risks of AI
- LG 1-4: Understand and explain the differences between AI systems and traditional software and decide on the appropriate type of solution
- LG 1-5: Know roles, their responsibilities, and how they collaborate in the context of AI
- LG 1-6 [OPTIONAL]: Know the concept of the Productivity J-Curve as a framework for introducing AI in organizations
- LG 2-1: Know how data protection laws affect the implementation and use of AI
- LG 2-2: Analyze the requirements of the EU AI Act and derive architecture-relevant measures for data flows, logging, technical documentation, human oversight, robustness, and deployment
- LG 2-3: Classify AI systems according to the EU AI Act risk levels
- LG 2-4: Evaluate copyright, degrees of openness, and licensing terms for AI models and AI-generated content
- LG 2-5: Define strategies for complying with regulatory requirements and specify suitable evidence and documentation artifacts for models, datasets, and operational processes
- LG 2-6: Know security pitfalls and types of attacks on ML models and select suitable security measures against attacks on AI systems
- LG 2-7: Know and apply strategies for AI risk mitigation
- LG 2-8: Understand the fundamental issues and facets of AI safety
- LG 2-9: Translate ethical guidelines and AI governance principles into organizational, development, and architectural decisions
- LG 3-1: Understand the life cycle of a machine learning or data science project
- LG 3-2: Understand data types and requirements as well as typical ML problems and their requirements
- LG 3-3: Select and compare suitable architecture and design patterns for AI systems and justify the selection based on quality requirements
- LG 3-4: Know metrics for measuring the performance of ML models
- LG 3-5: Integrate AI components into existing systems and domain architectures using suitable integration patterns, interfaces, and decoupling mechanisms
- LG 3-6: Design user interactions for AI systems that appropriately account for uncertainty, explainability, approvals, and human-in-the-loop
- LG 3-7: Analyze and design the performance and scalability of AI systems
- LG 3-8: Understand robustness in AI systems and apply strategies to increase robustness
- LG 3-9: Classify the reliability and availability of AI systems
- LG 3-10: Specify measures for the reproducibility, auditability, and traceability of models, dataset versions, prompts, configurations, and results

- LG 3-11: Classify explainability and interpretability in AI systems
- LG 3-12: Know fault tolerance in AI systems
- LG 3-13: Document architectural decisions for AI systems
- LG 4-1: Acquire and label data
- LG 4-2: Evaluate the sources and suitability of publicly available datasets in terms of quality, licensing, representativeness, and risks
- LG 4-3: Design data pipelines and data architectures for AI systems that support quality, traceability, scalability, governance, and reusability
- LG 4-4: Know strategies for data aggregation, cleansing, transformation, enrichment, and augmentation
- LG 4-5: Know options for storing data
- LG 4-6: Classify data products, data contracts, and responsibilities in data architectures for AI systems and use them to establish robust interfaces between the business domain, data platform, and AI components
- LG 5-1: Know (hardware) requirements for training and inference
- LG 5-2: Know the trade-offs of different model architectures with regard to quality characteristics
- LG 5-3: Adjust different quality features of an ML model
- LG 5-4: Understand the costs, power consumption, and sustainable use of AI (Green IT)
- LG 5-5: Gain an overview of MLOps for automating the life cycle of a data science project
- LG 5-6: Track model training, parameters, metrics, and results
- LG 5-7: Distinguish model evaluation, system evaluation, and assessment of business value, and select suitable evaluation approaches for AI systems
- LG 5-8: Know types of drift and possible causes and solutions for them
- LG 5-9: Gain an overview of CI/CD pipelines, model management, and deployment strategies for AI models
- LG 5-10: State the advantages and disadvantages of SaaS and self-hosting and make make-or-buy decisions based on quality requirements
- LG 5-11: Design monitoring and observability for AI systems, particularly for drift, quality degradation, costs, latency, security events, and governance-relevant evidence
- LG 5-12: Understand user feedback and methods and tools for collecting it
- LG 5-13: [OPTIONAL] Know classes of MLOps tools and platforms and essential criteria for selecting them
- LG 6-1: Gain an overview of AI integration levels
- LG 6-2: Integrate AI systems into the overall architecture of an IT landscape
- LG 6-3: Develop a fundamental understanding of generative AI and how it works
- LG 6-4: Know architecture patterns for using LLMs and evaluate their limitations, risks, and trade-offs, particularly for RAG, function calling, fine-tuning, assistance systems, and agents
- LG 6-5: Know and understand use cases for RAG (Retrieval-Augmented Generation) and selected RAG techniques

- LG 6-6: Know types of prompt engineering
- LG 6-7: Analyze and architecturally classify agentic workflows with regard to planning, tool use, reflection, state management, control mechanisms, and multi-agent collaboration
- LG 6-8: Know a selection of design patterns for generative AI systems
- LG 6-9: Select techniques for evaluating LLM applications and apply them to response quality, retrieval quality, hallucinations, safety, costs, and user value
- LG 6-10: Gain an overview of well-known LLMs and selection criteria
- LG 6-11: Understand the importance of cost management for GenAI applications
- LG 6-12: Know classes of libraries, interfaces, and tools for LLM applications as well as their application contexts and selection criteria
- LG 6-13: [OPTIONAL] Know and evaluate agentic AI architectures, key architecture components, security and governance boundaries, and typical usage patterns and limitations of agent systems
- LG 6-14: Architecturally classify and apply agent runtime systems for LLM and agent systems
- LG 7-1: Use an end-to-end case study or practical project to make, document, and justify architectural decisions for an AI system and evaluate them in terms of quality, compliance, operations, and integration.

Introduction: General information about the iSAQB Advanced Level

What is taught in an Advanced Level module?

The module can be attended independently of a CPSA-F certification.

- The iSAQB Advanced Level offers modular training in three areas of competence with flexible training paths. It takes into account individual inclinations and specializations.
- The certification takes the form of a term paper. The assessment and oral examination are carried out by experts appointed by the iSAQB.

What are Advanced Level (CPSA-A) graduates qualified for?

CPSA-A graduates can:

- design medium-sized to large IT systems independently and methodically
- assume technical and content-related responsibility in IT systems of medium to high criticality
- Conceptualize, design and document measures to achieve quality requirements and support development teams in the implementation of these measures
- Manage and carry out architecture-related communication in medium to large development teams

Requirements for CPSA-A certification

- Successful training and certification as a Certified Professional for Software Architecture, Foundation Level® (CPSA-F)
- At least three years of full-time professional experience in the IT sector; collaboration on the design and development of at least two different IT systems
 - Exceptions are allowed on application (e.g., collaboration on open source projects)
- Training and further education within the scope of iSAQB Advanced Level training courses with a minimum of 70 credit points from at least three different areas of competence
- Successful completion of the CPSA-A certification exam



Essentials

What does the module “SWARC4AI” convey?

The module presents participants with modern software architecture concepts for AI systems as a means of designing powerful, scalable and integrable AI solutions. At the end of the module, participants will be familiar with the key principles of software architecture for AI systems and will be able to apply them to the design and implementation of machine learning and generative AI systems. With the help of the modeling techniques and architecture tools taught, they will be able to seamlessly integrate AI components into existing software systems. The course covers both machine learning systems and generative AI and teaches how these can be combined with classic software systems. Participants learn what the architecture for such hybrid systems must look like in order to ensure scalability, maintainability and extensibility.

Curriculum Structure and Recommended Durations

Contents	Recommended minimum duration (min)
Introduction to concepts relevant to software architecture for artificial intelligence	120
Compliance, security, alignment	120
Design and development of AI systems	320
Data management and data processing for AI systems	90
Important quality features for the operation of AI systems	160
System architectures and platforms for generative AI systems	160
Case studies and practical projects	110
Total	1080

Duration, didactics and further details

The duration of a training course on the SWARC4AI module should be at least 3 days, but can be longer. Providers may differ in terms of duration, didactics, type and structure of the exercises and the detailed course structure. In particular, syllabus leaves the type of examples and exercises completely open. Licensed training courses on SWARC4AI contribute to admission to the final Advanced Level certification examination (following credit points):

Methodical Competence:	10 Points
Technical Competence:	20 Points
Communicative Competence:	0 Points

Prerequisites

Participants should have the following knowledge and/or experience:

- A basic understanding of artificial intelligence, machine learning and data science
- A basic understanding of software architecture, DevOps and the design of software systems and APIs
- A basic knowledge of the Python programming language and its use for AI problems
- Overview of common libraries such as scikit-learn, TensorFlow and PyTorch

Knowledge in the following areas can be helpful for understanding some concepts

- Experience with the command line interface on Linux systems
- Knowledge from the iSAQB CPSA Foundation level training for a general understanding of software architecture, design patterns and methodologies

Structure of the curriculum

The individual sections of the curriculum are described according to the following structure:

- **Terms/concepts:** Essential core terms of this topic.
- **Teaching/practice time:** Defines the minimum amount of teaching and practice time that must be spent on this topic or its practice in an accredited training course.
- **Learning goals:** Describes the content to be conveyed including its core terms and principles.

This section therefore also outlines the skills to be acquired in corresponding training courses.

Supplementary Information, Terms, Translations

To the extent necessary for understanding the curriculum, we have added definitions of technical terms to the [iSAQB glossary](#) and complemented them by adding references to the literature.

1. Introduction to software architecture-relevant concepts for artificial intelligence

Duration: 120 min	Practice time: 0 min
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1.1. Terms and Concepts

AI, Generative AI, Machine Learning, Symbolic AI, Evolutionary Algorithms, Statistical Learning, Deep Learning, LLMs, Hallucination, Bias, Jagged Technological Frontier, Data Preparation, Model Training, Model Evaluation, Feature Engineering, Data Science, Data Engineering

1.2. Learning Goals

LG 1-1: Classify artificial intelligence and machine learning, data science, deep learning, and generative AI

Participants know how artificial intelligence is defined and how machine learning, data science, deep learning, and generative AI fit within it.

LG 1-2: Identify, classify, prioritize, and evaluate AI use cases from a business perspective

Participants can identify and describe typical AI use cases from a business perspective. They can evaluate their suitability based on business goals, data availability and quality, technical feasibility, expected benefits, risks, integration effort, and impact on the software architecture, and can establish a well-founded prioritization.

Typical use cases include image recognition and image generation, natural language processing, predictive models, personalization, anomaly detection, chatbots, recommender systems, and intelligent assistance systems.

LG 1-3: Classify the strengths, limitations, and typical risks of AI

Participants can classify the strengths and limitations of AI technologies in different application contexts and explain why the suitability of AI depends heavily on the task at hand.

They can identify typical technical and societal risks, such as

- hallucinations,
- bias,
- unfairness,
- deepfakes,
- AI-assisted cyberattacks,
- safety risks in critical systems,
- social manipulation,
- infringements of intellectual property rights.

Participants are familiar with the “Jagged Technological Frontier” in relation to the task-specific suitability of AI.

LG 1-4: Understand and explain the differences between AI systems and traditional software and decide on the appropriate type of solution

Participants understand the differences between AI systems and traditional software:

- Data-driven (for ML) – data-centric rather than code-centric development
- Probabilistic results (non-deterministic behavior)
- Statistical validation
- Experimental design: support for rapid iteration and model testing
- Model complexity and interpretability: ML models, particularly deep learning models, can be extremely complex. Their “black-box” nature makes interpretation and explanation more difficult.
- Debugging and testing are more complex due to non-determinism.
- Model decay: continuous monitoring and retraining are necessary to maintain model performance.
- AI-specific regulations at industry and EU level must be taken into account.
- Interoperability: seamless integration into existing systems and technology stacks.

Participants can decide and explain whether and why a problem should be solved using AI or traditional software development.

LG 1-5: Know roles, their responsibilities, and how they collaborate in the context of AI

Participants know typical roles and their responsibilities in these contexts. In particular, these roles include:

Data Scientist, Data Analyst, Data Engineer, Machine Learning Engineer, MLOps Engineer, AI Architect, Data Architect, Business Intelligence (BI) Developer, Data Governance Specialist, and ML Researcher.

Participants also know how these roles can collaborate within a team (team topologies for ML teams).

LG 1-6 [OPTIONAL]: Know the concept of the Productivity J-Curve as a framework for introducing AI in organizations

This phenomenon helps explain why organizations may initially experience a decline in productivity when implementing AI, followed by productivity gains as implementation progresses.

1.3. References

[Agrawal et al.], [Bahree 2024], [Brynjolfsson et al.], [Burkov 2019], [Chong et al.], [Dell’Acqua 2026], [Géron 2022], [Gharbi et al. 2026], [Hall et al. 2023], [Harvard et al. 2024], [Huyen 2022], [Kelleher 2015], [Roser 2022], [Tan et al.], [Vaughan 2020], [Visengeriyeva, JTF], [Wang et al. 2024]

2. Compliance, security, alignment

Duration: 120 min	Practice time: 30 min
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2.1. Terms and Concepts

EU AI Act, data protection, copyright, license, open (source), AI security, jailbreak, adversarial attack, data poisoning, model inversion & extraction, AI safety, AI ethics, AI alignment, model and data documentation, transparency obligation, human oversight, data governance, AI governance, AI systems by risk levels (prohibited, high-risk, limited risk, low-risk)

2.2. Learning Goals

LG 2-1: Know how data protection laws affect the implementation and use of AI

Participants are familiar with data protection laws such as the GDPR and know how they affect the collection, processing, and storage of data by AI systems.

LG 2-2: Analyze the requirements of the EU AI Act and derive architecture-relevant measures for data flows, logging, technical documentation, human oversight, robustness, and deployment

Participants understand the objectives and provisions of the EU AI Act and know how they affect the development and use of AI systems. They also understand the EU AI Act requirements for trustworthy AI and how these requirements affect the development process and the architecture of the software system. In particular, they are familiar with the impact on some of the following aspects:

- Risk management system (risk mitigation)
- Data quality and data governance (quality management system)
- Creation and maintenance of comprehensive technical documentation for the AI system
- Automatic recording/logging of events in the AI system
- Provision of clear and comprehensible information to users
- Implementation of human oversight measures
- Assurance of an appropriate level of accuracy and robustness
- Implementation of cybersecurity measures

LG 2-3: Classify AI systems according to the EU AI Act risk levels

Participants know how AI systems are classified according to the EU AI Act risk levels (prohibited, high risk, limited risk, and low risk) and which regulatory requirements apply in each case.

LG 2-4: Evaluate copyright, degrees of openness, and licensing terms for AI models and AI-generated content

Participants can classify copyright issues relating to training data and AI-generated content and explain their impact on the development, provision, and use of AI systems.

They can distinguish different degrees of openness of ML models, for example with regard to training data, source code, model architecture, and model parameters, and evaluate licensing terms and the resulting restrictions on an AI system.

LG 2-5: Define strategies for complying with regulatory requirements and specify suitable evidence and documentation artifacts for models, datasets, and operational processes

Participants understand the fundamental provisions of the EU AI Act, particularly its transparency obligations, and know strategies for complying with them as well as potential challenges.

In particular, they can select documentation artifacts for models, datasets, prompts, evaluation procedures, changes, and operational processes, and explain how these support traceability, transparency, and auditability.

Participants can classify regulatory sandboxes as an instrument for the controlled testing of innovative AI systems and explain their opportunities, limitations, and the consequences of non-compliance with the EU AI Act.

LG 2-6: Know security pitfalls and types of attacks on ML models and select suitable security measures against attacks on AI systems

Participants know potential security pitfalls and typical types of attacks on ML models. These include, for example:

- LLM jailbreaks through prompt engineering
- Adversarial attacks
- Data poisoning
- Model inversion and extraction

Participants know security measures for protecting against attacks (AI security), particularly ways to integrate security standards and safeguards into the architecture, interfaces, and operating model.

LG 2-7: Know and apply strategies for AI risk mitigation

Participants know how strategies for AI risk mitigation can be developed and applied. In particular, they know the following strategies:

- Strengthening robustness through extensive testing
- Fault-tolerant AI systems
- Transparent development
- Explainable AI (XAI)

LG 2-8: Understand the fundamental issues and facets of AI safety

Participants understand the fundamental issues of AI safety and are familiar with its various facets. In particular, they know specific problems such as “AI model risks by poisoning” and bias.

Participants know how bias in data and models can be detected and reduced to ensure fairness and equal treatment in AI applications.

LG 2-9: Translate ethical guidelines and AI governance principles into organizational, development, and architectural decisions

Participants are aware of the ethical issues that AI systems can create and know approaches and options for dealing with ethical problems, for example by creating AI guidelines. They have an overview of important ethical guidelines such as the “Ethics Guidelines for Trustworthy AI” of the EU.

Optionally, participants know important documents and frameworks on AI governance that can be used to develop the core principles of AI governance and responsible AI for an organization, for example:

- OECD AI Principles, <https://oecd.ai/en/ai-principles>
- The Asilomar AI Principles, <https://futureoflife.org/open-letter/ai-principles/>
- The IEEE Ethically Aligned Design framework, https://standards.ieee.org/wp-content/uploads/import/documents/other/ead_v2.pdf

2.3. References

[ATLAS], [Bhajaria 2022], [Chen et al. 2022], [CSIRO et al. 2023], [EU AI Act], [Gharbi et al. 2026], [Hall et al. 2023], [Hotz, Best Practices], [Jarmul 2023], [Masood et al. 2023], [Molnar 2024], [Nist], [Pruksachatkun et al. 2023], [Visengeriyeva, Ethics]

3. Design and development of AI systems

Duration: 320 min	Practice time: 45 min
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3.1. Terms and Concepts

CI/CD, MLOps, CRISP-ML(Q), AI application, AI engineering, ML model development, ML infrastructure, performance, robustness, reliability, fault tolerance, model-as-service, model-as-dependency, precompute, model-on-demand, hybrid serving, multi-agent system (MAS)

3.2. Learning Goals

LG 3-1: Understand the life cycle of a machine learning or data science project

Participants have an understanding of the life cycle of a machine learning or data science project. In particular, they are familiar with the following phases:

- Exploratory data analysis
- Data cleansing and preparation
- Feature engineering
- Model training and selection
- POC
- Deployment
- Maintenance

Participants know typical process models for the software development of AI systems. These include, for example:

- CRISP-ML(Q)
- Team Data Science Process
- GenAI Life Cycle

LG 3-2: Understand data types and requirements as well as typical ML problems and their requirements

Participants know different types of data as well as typical ML problems and use cases involving data from the following areas:

- Supervised learning
- Unsupervised learning
- Reinforcement learning

They understand the requirements these may have, for example:

- Availability of correct labels of different types
- Data volume and diversity
- Temporal or semantic dependencies in data features

- Encoding the input data of ML models, for example:
 - One-hot encodings
 - Label encodings
 - Embeddings

In particular, participants have a good understanding of the need for validation and of the typical division of data into training, validation, and test datasets.

LG 3-3: Select and compare suitable architecture and design patterns for AI systems and justify the selection based on quality requirements

Participants know which design patterns exist for AI systems and how to select suitable patterns. This includes the following design patterns:

- ML systems topology patterns
- Pipeline architecture patterns
- Model training patterns
- Model serving patterns
- Model deployment patterns

In particular, participants know transfer learning and fine-tuning as ways to adapt pretrained foundation models to existing use cases.

LG 3-4: Know metrics for measuring the performance of ML models

Participants are familiar with various metrics for measuring the performance of ML models, such as:

- Precision, recall
- F1 score
- Accuracy

and know how to define evaluation criteria for performance assessment.

LG 3-5: Integrate AI components into existing systems and domain architectures using suitable integration patterns, interfaces, and decoupling mechanisms

Participants understand how ML models can be integrated into existing systems and are familiar with the interfaces and integration points.

LG 3-6: Design user interactions for AI systems that appropriately account for uncertainty, explainability, approvals, and human-in-the-loop

Participants know how user interfaces should be designed to enable effective interactions with the ML system and optimize the user experience.

LG 3-7: Analyze and design the performance and scalability of AI systems

Participants understand the importance of performance indicators such as latency, throughput, and resource utilization in AI systems for both training and inference, and understand how these can be optimized.

They can analyze the causes of performance and scalability problems and select suitable measures that enable AI systems to handle growing data volumes, numbers of users, and request volumes.

LG 3-8: Understand robustness in AI systems and apply strategies to increase robustness

Participants have an understanding of what robustness means in AI systems and can apply strategies to increase robustness in different application contexts.

LG 3-9: Classify the reliability and availability of AI systems

Participants understand the concepts of reliability and availability and know how to build AI systems that are stable and constantly available.

LG 3-10: Specify measures for the reproducibility, auditability, and traceability of models, dataset versions, prompts, configurations, and results

Participants know the importance of AI results being reproducible and auditable and know which methods can be used to ensure these properties.

LG 3-11: Classify explainability and interpretability in AI systems

Participants understand the importance of explainability and interpretability in AI systems and know how to ensure this in order to promote trust and transparency.

LG 3-12: Know fault tolerance in AI systems

Participants are familiar with the concepts of fault tolerance and can explain how AI systems remain functional despite errors or malfunctions.

LG 3-13: Document architectural decisions for AI systems

Participants know how to define and document the use cases or tasks of an ML model, such as image classification or fraud detection, particularly the trade-offs between:

- RAG versus fine-tuning
- SaaS versus self-hosting
- Batch versus real time
- A centralized versus a domain-specific AI service

3.3. References

[Bornstein et al.], [Cdteliot], [Crowe et al. 2024], [Gharbi et al. 2026], [Heiland et al. 2023], [Hotz, Life Cycle], [Hotz, TDSP], [Koc], [Lakshmanan et al.], [ML software architecture], [Nahar et al.], [Saltz], [Savarese], [Serban], [Studer et al.], [tdcox], [TU Berlin], [Visengeriyeva, AI Agents], [Zaharia et al.]

4. Data management and data processing for AI systems

Duration: 90 min	Practice time: 0 min
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4.1. Terms and Concepts

Data acquisition, labeling, data pipeline, ETL processes, data aggregation, data cleansing, transformation, augmentation, files, RDBMS, NoSQL, data products, data contracts, data architectures: Data Warehouse, Data Lake, Data Mesh

4.2. Learning Goals

LG 4-1: Acquire and label data

Participants have an overview of various methods for acquiring and labeling data.

Participants know relevant data-labeling tools such as CVAT, Label Studio, or Amazon Mechanical Turk.

LG 4-2: Evaluate the sources and suitability of publicly available datasets in terms of quality, licensing, representativeness, and risks

Participants can name common platforms for publicly accessible data.

LG 4-3: Design data pipelines and data architectures for AI systems that support quality, traceability, scalability, governance, and reusability

Participants understand the design of efficient data pipelines and architectures, the consideration of data quality, and storage solutions and their management. To this end, participants know:

- Architecture patterns for data engineering pipelines and ETL processes
- How effective data management ensures the quality and security of data in AI applications

Participants have an overview of relevant tools for data engineering pipelines, such as Apache Spark and Flink.

LG 4-4: Know strategies for data aggregation, cleansing, transformation, enrichment, and augmentation

Participants know strategies for data aggregation, cleansing, transformation, enrichment, and augmentation.

LG 4-5: Know options for storing data

Participants know various options for storing data as well as their advantages and disadvantages. The options include the following technologies:

- CSV files
- Column-oriented files
- Relational and document databases
- Data warehouses
- Data lakes

- Vector databases
- Graph databases

LG 4-6: Classify data products, data contracts, and responsibilities in data architectures for AI systems and use them to establish robust interfaces between the business domain, data platform, and AI components

Participants know fundamental concepts of modern data architectures for AI systems and understand the role of data products, data contracts, and business and technical responsibilities in stable and scalable data flows.

4.3. References

[Bhajaria 2022], [Bornstein et al.], [Dehghani], [Ford et al.], [Gharbi et al. 2026], [Jones], [Reis et al.], [Sanderson et al.], [Sarkis], [Serra]

5. Important quality features for the operation of AI systems

Duration: 160 min	Practice time: 30 min
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5.1. Terms and Concepts

Quality features, scalability, performance optimization, monitoring, logging, feedback, FinOps for AI platforms

5.2. Learning Goals

LG 5-1: Know (hardware) requirements for training and inference

Participants know the different (hardware) requirements for training and inference, for example for TPUs, NPUs, LPUs, GPUs, or CPUs.

LG 5-2: Know the trade-offs of different model architectures with regard to quality characteristics

Participants can give examples of trade-offs between different model architectures with regard to quality characteristics. Particularly for scaling, efficiency, and memory footprint, participants should know the trade-offs as well as the advantages and disadvantages of important architectures such as RNNs and transformers.

LG 5-3: Adjust different quality features of an ML model

Participants know ways to balance and trade off different quality characteristics of an ML model, such as accuracy, efficiency, and memory footprint. In particular, they know the following techniques:

- Quantization
- Pruning
- Distillation
- LoRA

LG 5-4: Understand the costs, power consumption, and sustainable use of AI (Green IT)

Participants gain an understanding of the costs, power consumption, and sustainable use of AI (Green IT). In particular, they know how to develop AI models and systems that operate efficiently by making economical use of memory, compute capacity, and storage space.

LG 5-5: Gain an overview of MLOps for automating the life cycle of a data science project

Participants know the term MLOps for automating the life cycle of a data science project and its relationship to DevOps.

Using a practical example, participants understand what an MLOps pipeline can look like and what insights it provides into parameters, metrics, and other aspects.

LG 5-6: Track model training, parameters, metrics, and results

Participants understand tracking in model training, including parameters, metrics, and results.

LG 5-7: Distinguish model evaluation, system evaluation, and assessment of business value, and select suitable evaluation approaches for AI systems

Participants have an overview of approaches for evaluating ML models and the AI systems based on them.

LG 5-8: Know types of drift and possible causes and solutions for them

Participants are familiar with different types of drift, such as data drift or model drift, as well as possible causes and solutions.

LG 5-9: Gain an overview of CI/CD pipelines, model management, and deployment strategies for AI models

Participants understand CI/CD pipelines, model management, and deployment strategies for AI models.

Participants know a selection of different deployment options for AI models:

- API deployment
- Embedded deployment
- Batch prediction
- Streaming
- Containerization
- Serverless deployment
- Cloud services

Participants know common platforms and tools for model provision and for creating POCs and understand how they work conceptually.

LG 5-10: State the advantages and disadvantages of SaaS and self-hosting and make make-or-buy decisions based on quality requirements

Participants know the advantages and disadvantages of SaaS and self-hosting and can weigh them against each other.

Participants can make make-or-buy decisions for MLOps and AI platform systems or components based on quality requirements, cost, risk, compliance, and integration effort.

Participants have an overview of well-known SaaS AI solutions, such as Azure OpenAI Services.

LG 5-11: Design monitoring and observability for AI systems, particularly for drift, quality degradation, costs, latency, security events, and governance-relevant evidence

Participants understand the need for monitoring, including with regard to AI-specific requirements such as drift tracking.

Participants know relevant metrics such as accuracy, precision, recall, F1 score, MAE, MSE, perplexity, latency, throughput, and resource utilization, and understand why they are relevant for monitoring.

Participants know example monitoring tools. These include both general-purpose tools, such as Prometheus and Grafana, and ML-specific tools, such as MLflow.

LG 5-12: Understand user feedback and methods and tools for collecting it

Participants understand the benefits of user feedback for further model training. They also know various methods and tools for collecting user feedback, such as choosing between multiple answers or flagging in

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Participants know further methods for obtaining user feedback, such as RLHF, RLAI, and DPO.

LG 5-13: [OPTIONAL] Know classes of MLOps tools and platforms and essential criteria for selecting them

Participants know well-known MLOps tools and end-to-end platforms, such as:

- Domino Data Lab, h2o.ai, DVC, activeloop, aporia, argo, arize, bentoml, comet ML, DagsHub, Databricks MLOps Stacks, Feast, Kedro, Metaflow, MLflow, MLRun, prefect, PrimeHub, Weights & Biases, WhyLabs, zenML, KNIME, RapidMiner, NVIDIA AI Enterprise, watsonx.ai
- Open source: MLflow, Kubeflow, ClearML
- PaaS: AWS SageMaker, Azure ML, Google Vertex AI

5.3. References

[Chen et al. 2022], [Gharbi et al. 2026], [Haviv et al. 2023], [Kumara et al.], [Osipov 2022], [Salama et al.], [Tan Wei Hao et al. 2024], [Treveil et al. 2020], [Wilson 2022]

6. System architectures and platforms for generative AI systems

Duration: 160 min	Practice time: 30 min
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6.1. Terms and Concepts

Generative AI, LLMs, MLflow, Managed MLflow, Azure Machine Learning, Metaflow, Generative AI, LLM, (Stable) Diffusion, Vector DB, Embedding, RNN, Transformer, RAG, Agentic Workflows etc.

6.2. Learning Goals

LG 6-1: Gain an overview of AI integration levels

Participants are familiar with the various levels of AI integration. These include the following levels:

- Applications (e.g., coding assistants)
- AI engineering (e.g., prompt engineering)
- ML model development (e.g., PyTorch)
- ML infrastructure (e.g., vector databases)

Participants know some examples of common libraries, interfaces, and tools for integrating AI models.

LG 6-2: Integrate AI systems into the overall architecture of an IT landscape

Participants know how to integrate AI systems into IT landscapes at a strategic level. For example, they can use strategic design from DDD, particularly context maps, to determine and document the type and degree of integration of AI systems.

Using a case study with a fictional business domain, participants practice and discuss how to weigh integration options for AI in an existing software landscape.

LG 6-3: Develop a fundamental understanding of generative AI and how it works

Participants have a basic understanding of generative AI, such as LLMs and diffusion models.

Participants understand how LLMs work and can classify the associated terminology: token, embedding, RNN, transformer, attention.

LG 6-4: Know architecture patterns for using LLMs and evaluate their limitations, risks, and trade-offs, particularly for RAG, function calling, fine-tuning, assistance systems, and agents

Participants know established patterns for using LLMs. These include the following patterns:

- RAG and retrieval strategies
- Function calling
- Fine-tuning
- Assistants
- Agents

LG 6-5: Know and understand use cases for RAG (Retrieval-Augmented Generation) and selected RAG techniques

Participants know typical use cases for RAG, such as “Talk to your documents/database/API.”

In particular, participants know a selection of common RAG techniques, such as:

- Regular RAG, GraphRAG, CAG, agentic RAG, and multi-agent RAG
- Context enrichment techniques
- Fusion retrieval
- Intelligent reranking
- Query transformations
- Adaptive retrieval
- Iterative retrieval
- Ensemble retrieval
- Knowledge graph integration

LG 6-6: Know types of prompt engineering

Participants know different types of prompt engineering, such as:

- Few-shot learning
- Chain-of-thought
- Role-playing

as well as general best practices for prompting.

LG 6-7: Analyze and architecturally classify agentic workflows with regard to planning, tool use, reflection, state management, control mechanisms, and multi-agent collaboration

Participants know what agentic workflows are and are familiar with the concepts of reflection, tool use, planning, and multi-agent collaboration.

LG 6-8: Know a selection of design patterns for generative AI systems

Participants know which design patterns exist for generative AI systems. They know a selection of the following patterns:

- AI Query Router (Simple Router; Ranking-based Router; Learning-based Router)
- Layered Caching Strategy Leading to Fine-Tuning
- Multiplexing AI Agents
- Fine-Tuning LLMs for Multiple Tasks
- Blending Rules-Based and Generative Approaches
- Utilizing Knowledge Graphs with LLMs
- Swarm of Generative AI Agents
- Modular Monolith LLM Approach with Composability
- Memory Cognition for LLMs

- Red and Blue Team Dual-Model Evaluation

LG 6-9: Select techniques for evaluating LLM applications and apply them to response quality, retrieval quality, hallucinations, safety, costs, and user value

Participants know several techniques for evaluating LLM applications. These may include the following techniques, for example:

- Scoring
- Human feedback
- Comparative evaluation
- Model-based evaluation

Participants know common evaluation frameworks such as LangSmith or LangFuse for dealing with indeterminacy and errors in AI systems.

LG 6-10: Gain an overview of well-known LLMs and selection criteria

Participants are familiar with well-known LLMs such as GPT, Claude, Gemini, Llama, Mistral, or Luminous, and with criteria for selecting a suitable LLM.

LG 6-11: Understand the importance of cost management for GenAI applications

Participants understand the importance of cost management for GenAI applications.

LG 6-12: Know classes of libraries, interfaces, and tools for LLM applications as well as their application contexts and selection criteria

Participants know some examples of common libraries, interfaces, and tools related to LLM applications, such as the OpenAI API or LangChain.

LG 6-13: [OPTIONAL] Know and evaluate agentic AI architectures, key architecture components, security and governance boundaries, and typical usage patterns and limitations of agent systems

Participants know agentic AI software architectures, AI agent architecture components, and types of AI agent architectures.

LG 6-14: Architecturally classify and apply agent runtime systems for LLM and agent systems

Participants understand agent runtime systems as an architectural control and integration layer for probabilistic AI components such as LLMs and agents. They can describe harness approaches using established architecture concepts such as encapsulation, interface control, invariants, operational boundaries, observability, policy enforcement, fault isolation, controlled change, governance, and auditability, and can apply these approaches when designing production-ready AI systems.

In particular, they can derive how an agent runtime system:

- limits the autonomy of an AI component through clear interfaces, permitted actions, and operational boundaries,
- controls inputs, context, tool access, state, and outputs,
- enforces system invariants and policies,
- isolates errors and unsafe model outputs,

- ensures observability, traceability, and auditability,
- enables controlled changes to prompts, models, retrieval components, and tools,
- supports the secure operation and safe evolution of AI systems.

6.3. References

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7. Case studies and practical projects

Duration: 110 min	Practice time: 110 min
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7.1. Learning Goals

LG 7-1: Use an end-to-end case study or practical project to make, document, and justify architectural decisions for an AI system and evaluate them in terms of quality, compliance, operations, and integration.

7.2. References

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